SKILL - 7 REST API - CRUD Operations using ResponseEntity

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Skill Experiment Number: 07

Section Number:75

Code:

Skill7Application.java

```
package com.example.courseapi;


import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;


@SpringBootApplication
public class Skill7Application {


	public static void main(String[] args) {
		SpringApplication.run(Skill7Application.class, args);
	}


}
```

CourseController.java

```
package com.example.courseapi.controller;


import org.springframework.http.*;
import org.springframework.web.bind.annotation.*;
```

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```java
import java.util.*;

import com.example.courseapi.model.Course;
import com.example.courseapi.service.CourseService;

@RestController
@RequestMapping("/courses")
public class CourseController {

  private final CourseService service;

  public CourseController(CourseService service) {
    this.service = service;
  }

  @PostMapping
  public ResponseEntity<?> addCourse(@RequestBody Course course) {
    Course saved = service.addCourse(course);
    return ResponseEntity.status(HttpStatus.CREATED).body(saved);
  }

  @GetMapping
  public ResponseEntity<?> getCourses() {
    return ResponseEntity.ok(service.getAllCourses());
  }
```

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```
@PutMapping("/{id}")
public ResponseEntity<?> updateCourse(@PathVariable int id, @RequestBody Course course) {
    Course updated = service.updateCourse(id, course);

    if (updated == null) {
        return ResponseEntity.status(HttpStatus.NOT_FOUND)
                .body("Course not found");
    }

    return ResponseEntity.ok(updated);
}

@DeleteMapping("/{id}")
public ResponseEntity<?> deleteCourse(@PathVariable int id) {

    boolean deleted = service.deleteCourse(id);

    if (!deleted) {
        return ResponseEntity.status(HttpStatus.NOT_FOUND)
                .body("Course not found");
    }

    return ResponseEntity.ok("Course deleted successfully");
}
```

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```
    @GetMapping("/search/{title}")
    public ResponseEntity<?> searchCourse(@PathVariable String title) {

        List<Course> result = service.searchByTitle(title);

        if (result.isEmpty()) {
            return ResponseEntity.status(HttpStatus.NOT_FOUND)
                    .body("No courses found");
        }

        return ResponseEntity.ok(result);
    }
}
```

Course.java

```
package com.example.courseapi.model;

public class Course {

    private int courseId;
    private String title;
    private String duration;
    private double fee;

    public Course() {}

    public Course(int courseId, String title, String duration, double fee) {
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```
    this.courseId = courseId;
    this.title = title;
    this.duration = duration;
    this.fee = fee;
}

public int getCourseId() {
    return courseId;
}

public void setCourseId(int courseId) {
    this.courseId = courseId;
}

public String getTitle() {
    return title;
}

public void setTitle(String title) {
    this.title = title;
}

public String getDuration() {
    return duration;
}
```

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```
    public void setDuration(String duration) {
        this.duration = duration;
    }

    public double getFee() {
        return fee;
    }

    public void setFee(double fee) {
        this.fee = fee;
    }
}
```

CourseService.java

```
package com.example.courseapi.service;

import org.springframework.stereotype.Service;
import java.util.*;
import com.example.courseapi.model.Course;

@Service
public class CourseService {

    List<Course> courseList = new ArrayList<>();

    public List<Course> getAllCourses() {
        return courseList;
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```
}

public Course addCourse(Course course) {
    courseList.add(course);
    return course;
}

public Course updateCourse(int id, Course course) {
    for (Course c : courseList) {
        if (c.getCourseId() == id) {
            c.setTitle(course.getTitle());
            c.setDuration(course.getDuration());
            c.setFee(course.getFee());
            return c;
        }
    }
    return null;
}

public boolean deleteCourse(int id) {
    return courseList.removeIf(c -> c.getCourseId() == id);
}

public List<Course> searchByTitle(String title) {
    List<Course> result = new ArrayList<>();
    for (Course c : courseList) {
```

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```
        if (c.getTitle().equalsIgnoreCase(title)) {
            result.add(c);
        }
    }
    return result;
  }
}
```

Output:

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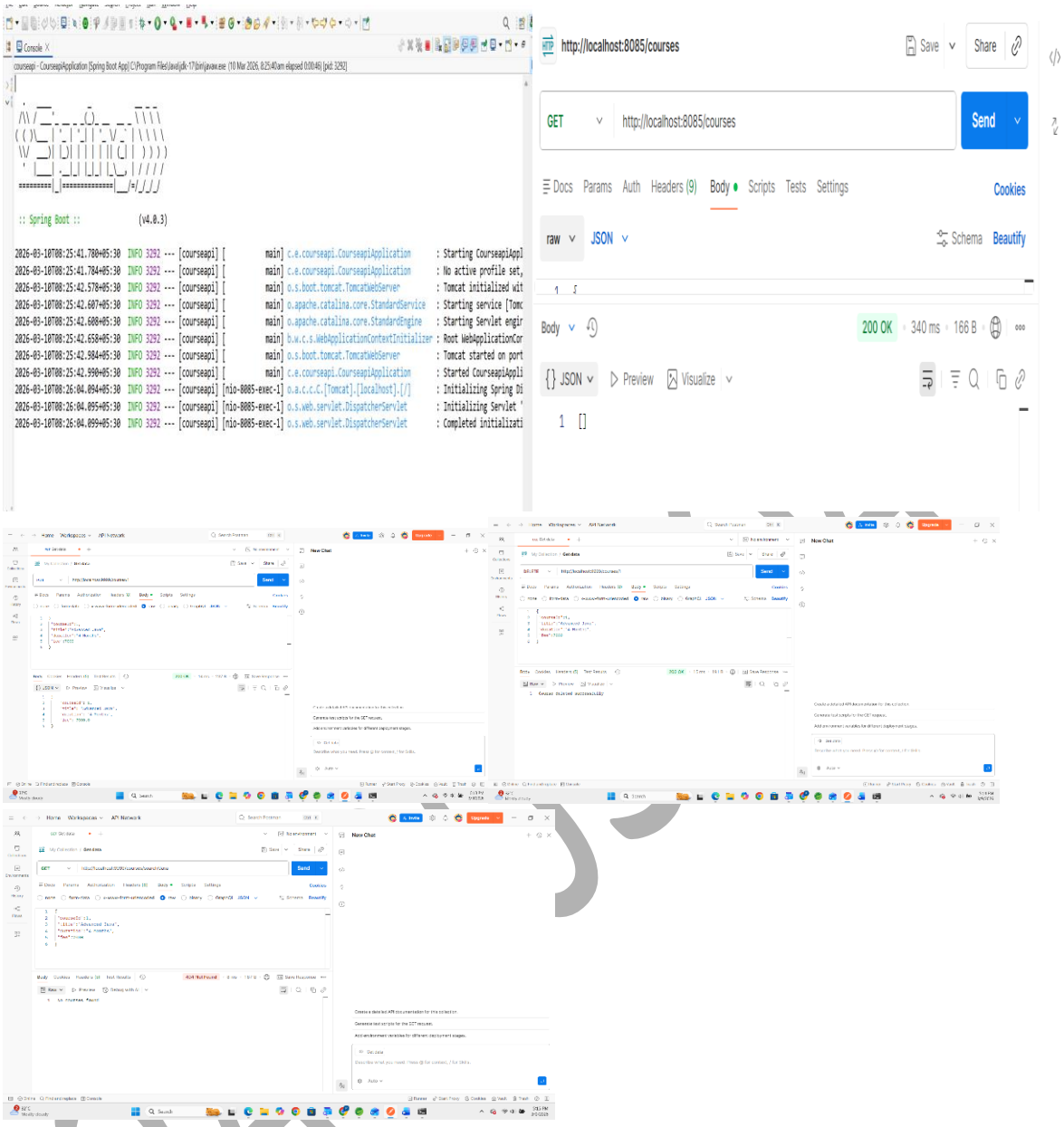

https://github.com/Gowtham-406/S-75_1041_SkillLabDocuments.git